BINUS University

School of Computer Science

Assignment: Cloud Resource Management & Scheduling

Course: Cloud Infrastructure

Instructions:

1. This is an individual assignment.

2. Answer all questions clearly and concisely.

3. Provide diagrams where necessary.

4. Submission in PDF format.

Case Scenario:

A video streaming platform is experiencing rapid user growth. During peak hours, the system suffers from high latency and resource contention. The platform runs on cloud infrastructure and needs better resource management and scheduling strategies.

Questions:

1. Resource Identification (20 points)

Identify and explain the key cloud resources required in this system (compute, storage, network). Describe the workload characteristics.

2. Problem Analysis (20 points)

Analyze at least three resource management issues that may occur in this system.

3. Scheduling Strategy Design (25 points)

Propose a scheduling strategy. Include load balancing, auto-scaling, and request handling.

4. Architecture Design (20 points)

Design a cloud architecture that supports efficient resource management.

5. Trade-off Analysis (15 points)

Discuss trade-offs between cost, performance, and scalability.